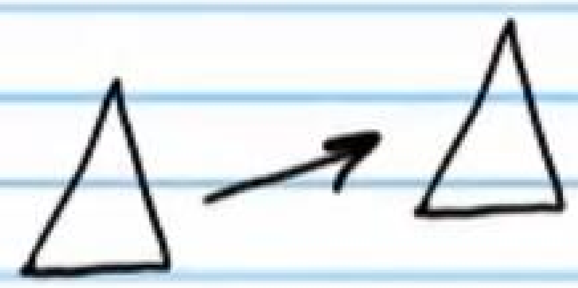
Name
Date
Course

Transformations and the Library of Relations (Part IV)

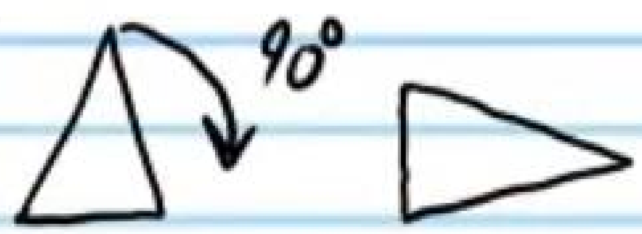
Review on Transformations

I) Translation.



II) Rotation.

* There must be a center of rotation



III) Reflection.

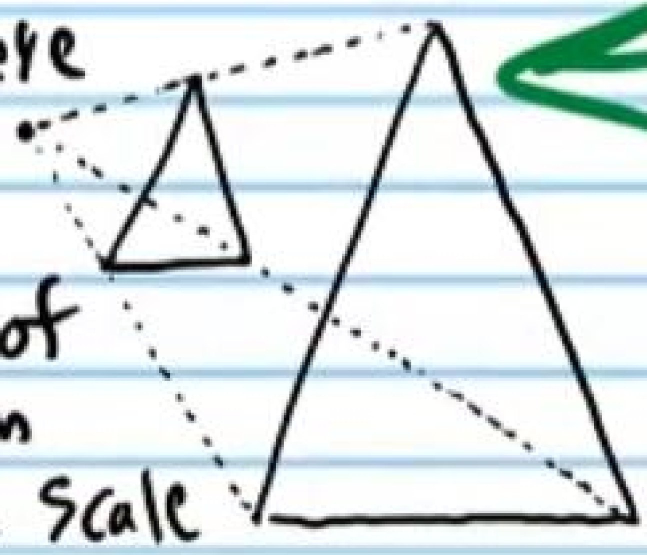
* There must be a line of reflection.

No Change in Size.

$\boxed{\text{Rigid}}$

IV) Dilation.

* There must be a center of dilation and a scale factor.



Size Changes.

$\boxed{\text{Non-Rigid}}$

Relations

Relation: A connection between two values (usually called X and Y).

Three Ways to Represent a Relation

Examples

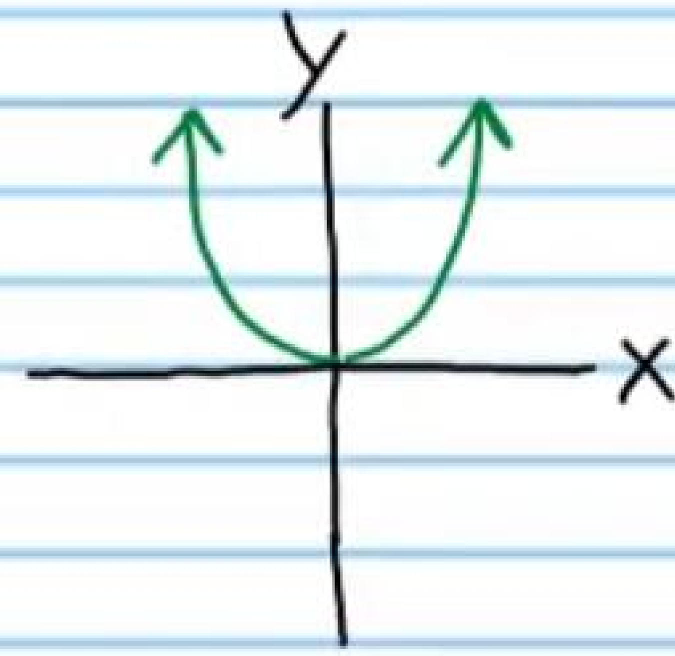
I. Equation $y = x^2$ $y = 3^x$

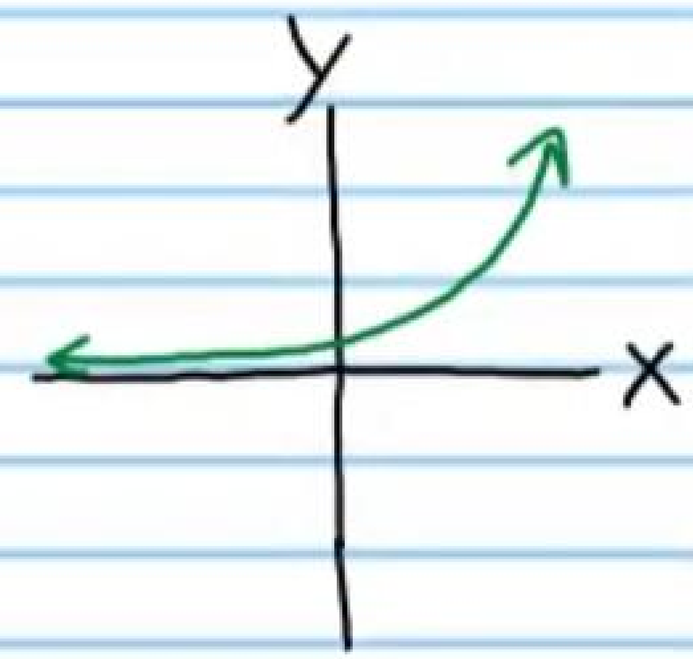
II. Table

X	-2	-1	0	1	2
Y	4	1	0	1	4

X	-2	-1	0	1	2
Y	1/9	1/3	1	3	9

III. Graph

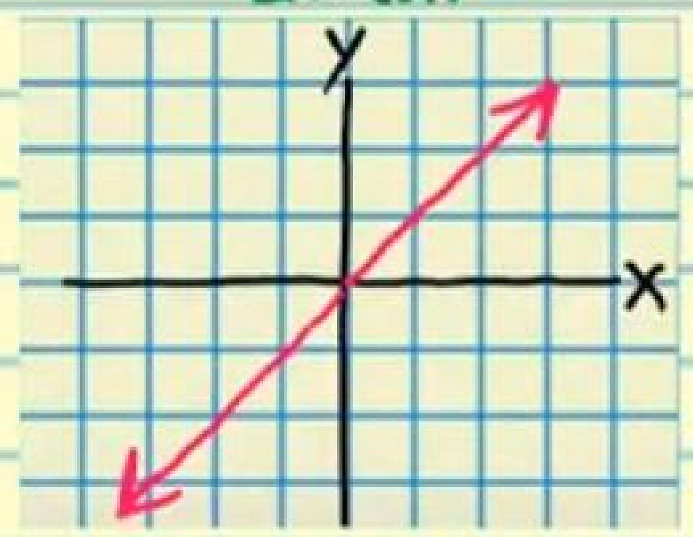




The Library of Relations

*Also called "parent" relations.

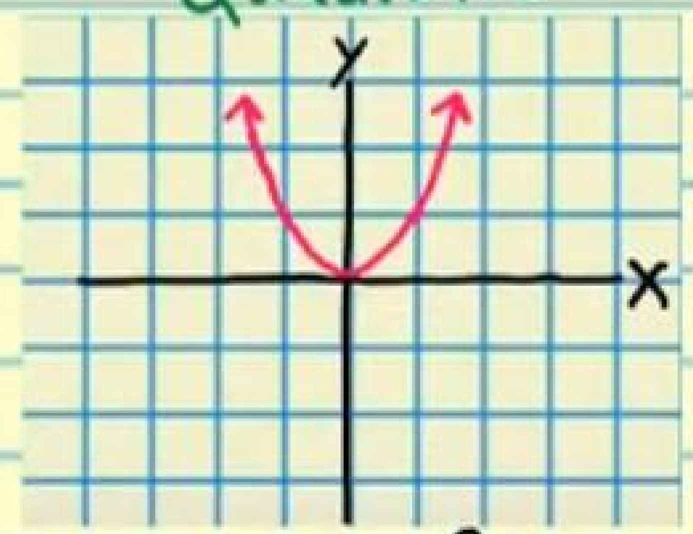
Linear



$y = x$

x	-2	-1	0	1	2
y	-2	-1	0	1	2

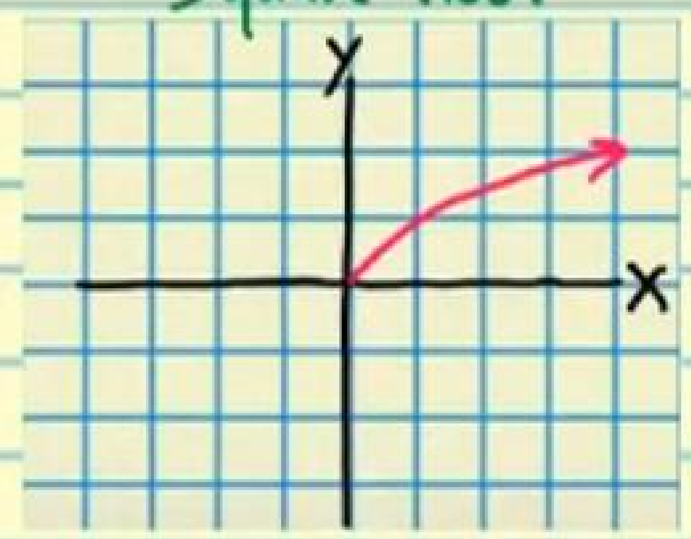
Quadratic



$y = x^2$

x	-2	-1	0	1	2
y	4	1	0	1	4

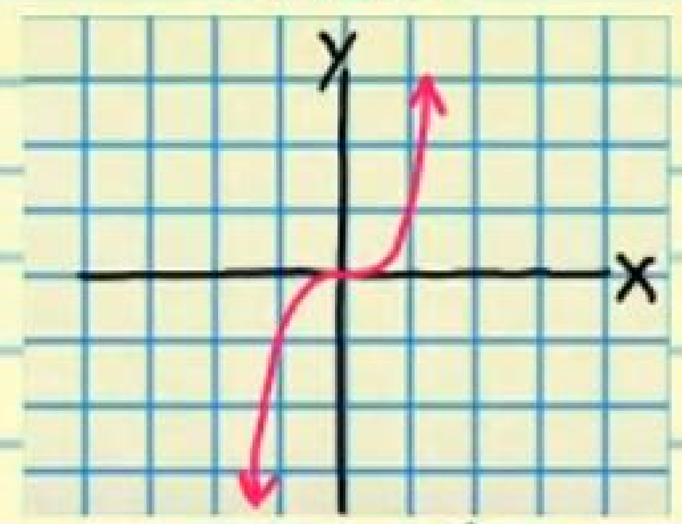
Square Root



$y = \sqrt{x}$

x	0	1	4	9	16
y	0	1	2	3	4

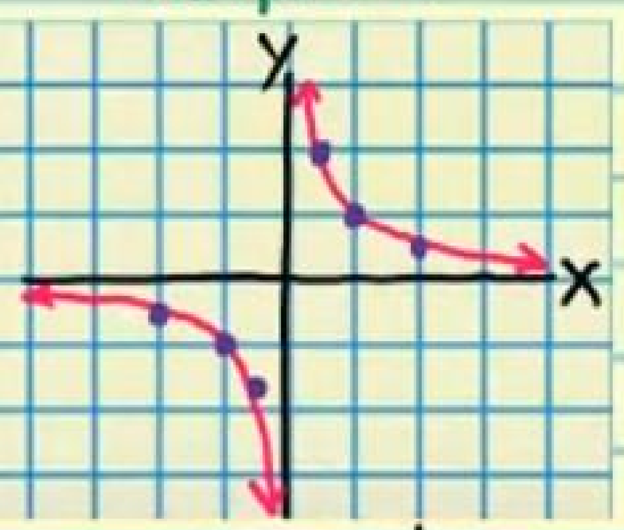
Cubic



$y = x^3$

x	-2	-1	0	1	2
y	-8	-1	0	1	8

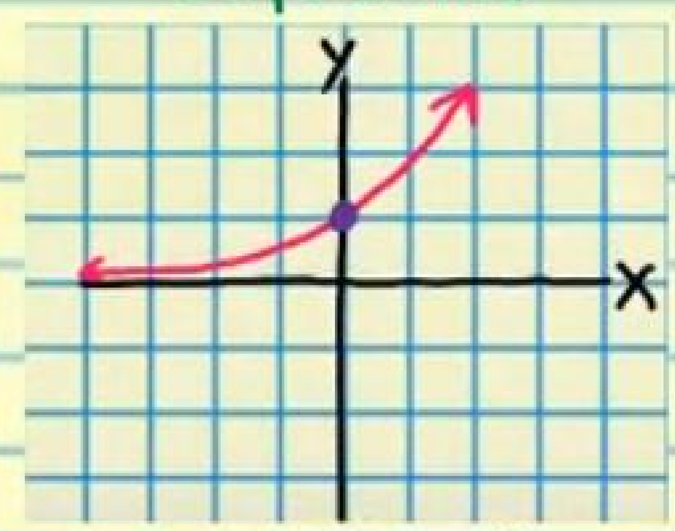
Reciprocal



$y = 1/x$

x	-2	-1	-1/2	0	1/2	1	2
y	-1/2	-1	-2	und	2	1	1/2

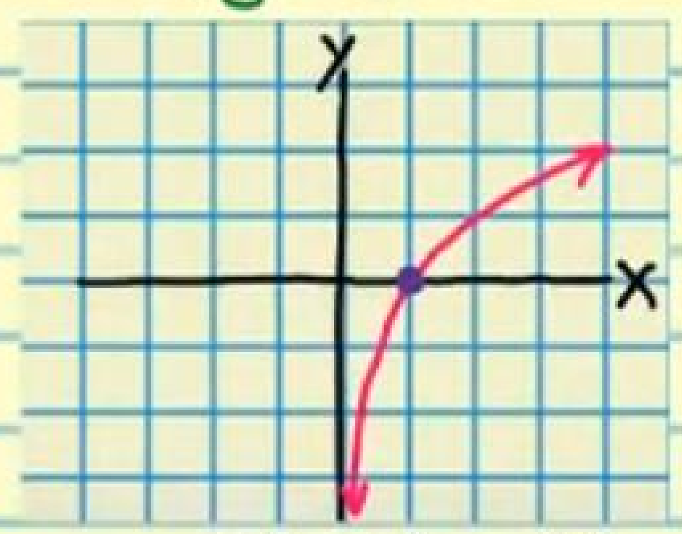
Exponential



$y = a^x, \ a > 1$

x	—	—	0	—	—
y	—	—	1	—	—

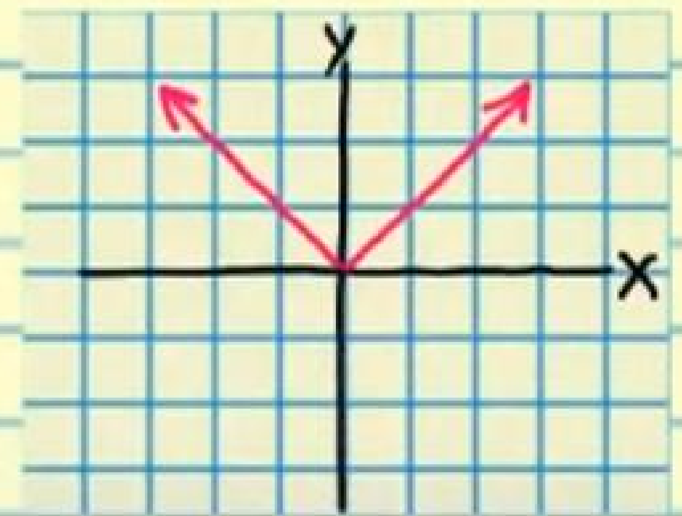
Logarithmic



$y = \log_a x$

x	0	1	—	—	—
y	und	0	—	—	—

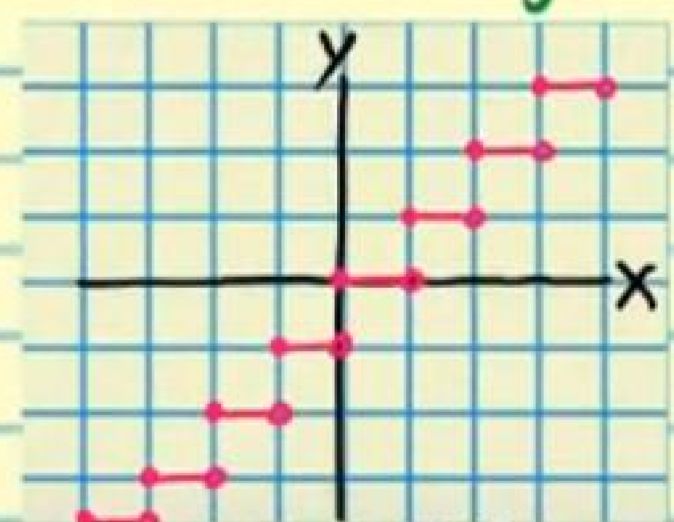
Absolute Value



$y = |x|$

x	-2	-1	0	1	2
y	2	1	0	1	2

Greatest Integer



$y = [x]$

x	-2 1/3	-1	0	1/2	1 1/2
y	-3	-1	0	0	1

The Transformation Properties

Translation

If y is replaced with $y+k$ in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units down (assuming that k is positive).

If y is replaced with $y-k$ in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units up (assuming that k is positive).

If x is replaced with $x+k$ in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units to the left (assuming that k is positive).

If x is replaced with $x-k$ in an equation, then the graph of the new equation is the same as the graph of the original equation, but moved k units to the right (assuming that k is positive).

Reflection

If y is replaced with $-y$ in an equation, then the graph of the new equation is the same as the graph of the original equation, but reflected across the x-axis.

If x is replaced with $-x$ in an equation, then the graph of the new equation is the same as the graph of the original equation, but reflected across the y-axis.

Dilation

Vertical Compression

If y is replaced with ky in an equation, then the graph of the new equation is the same as the graph of the original equation, except every y-value is divided by k (assuming that k is positive).

Vertical Stretch

If y is replaced with y/k in an equation, then the graph of the new equation is the same as the graph of the original equation, except every y-value is multiplied by k (assuming that k is positive).

Horizontal Compression

If x is replaced with kx in an equation, then the graph of the new equation is the same as the graph of the original equation, except every x-value is divided by k (assuming that k is positive).

Horizontal Stretch

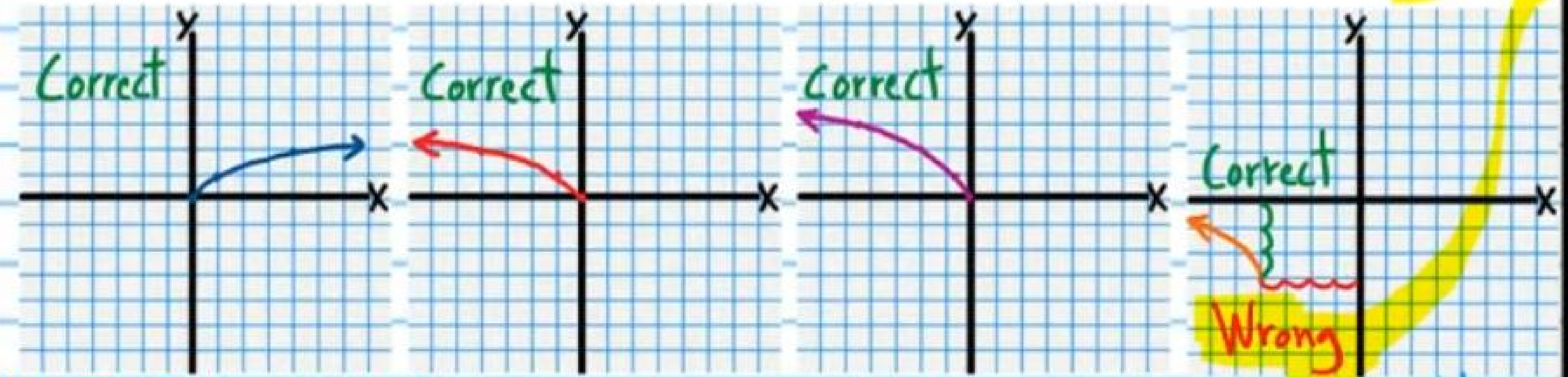
If x is replaced with x/k in an equation, then the graph of the new equation is the same as the graph of the original equation, except every x-value is multiplied by k (assuming that k is positive).

Learning to Use the Transformation Properties Correctly

Examples Demonstrating Incorrect Usage of Transformations

(1A) $y+3=\sqrt{-2x+4}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Reflection	Dilation	Translation?
$y=\sqrt{x}$	$y=\sqrt{-x}$	$y=\sqrt{-2x}$	$y+3=\sqrt{-2x+4}$

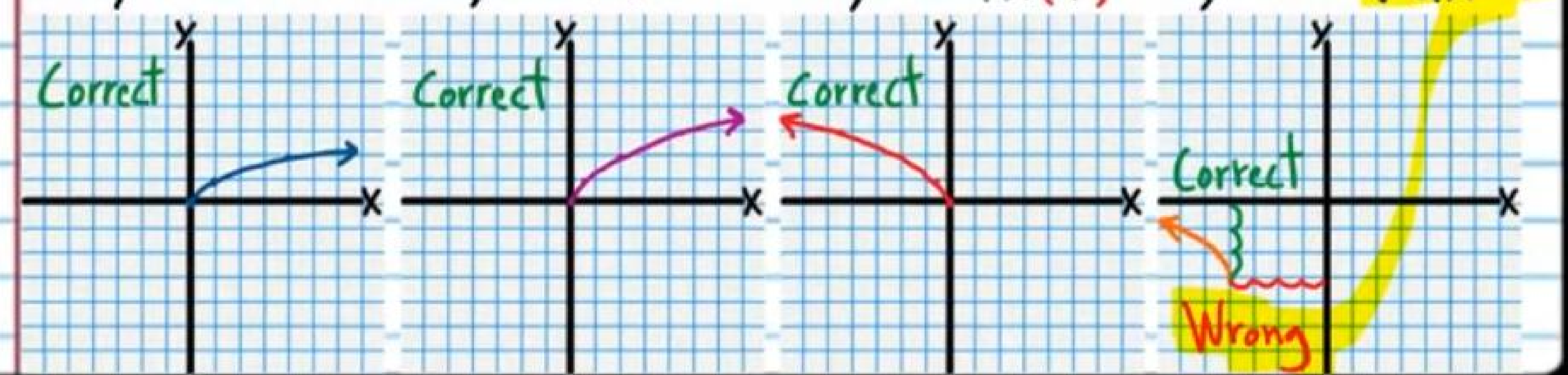


Why? When "x" is replaced with $x+4$ in an equation, the graph of the new equation is 4 units to the left of the graph of the original equation. However, we did not replace "x" with $x+4$. We replaced $-2x$ with $-2x+4$. If we replace "x" with $x+4$, we would get $\sqrt{-2(x+4)}$, which is equivalent to $\sqrt{-2x-8}$, but $-2x-8$ is not in the equation that we are trying to graph.

(1B) $y+3=\sqrt{-2x+4}$

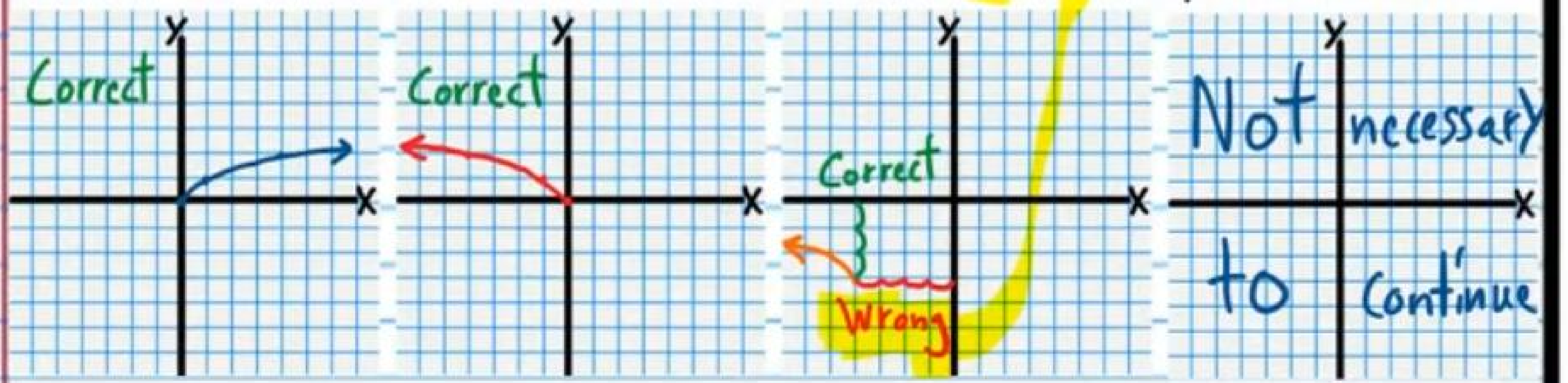
$2(-x)=-2x$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Dilation	Reflection	Translation?
$y=\sqrt{x}$	$y=\sqrt{2x}$	$y=\sqrt{2(-x)}$	$y+3=\sqrt{-2x+4}$



(1C) $y+3=\sqrt{-2x+4}$

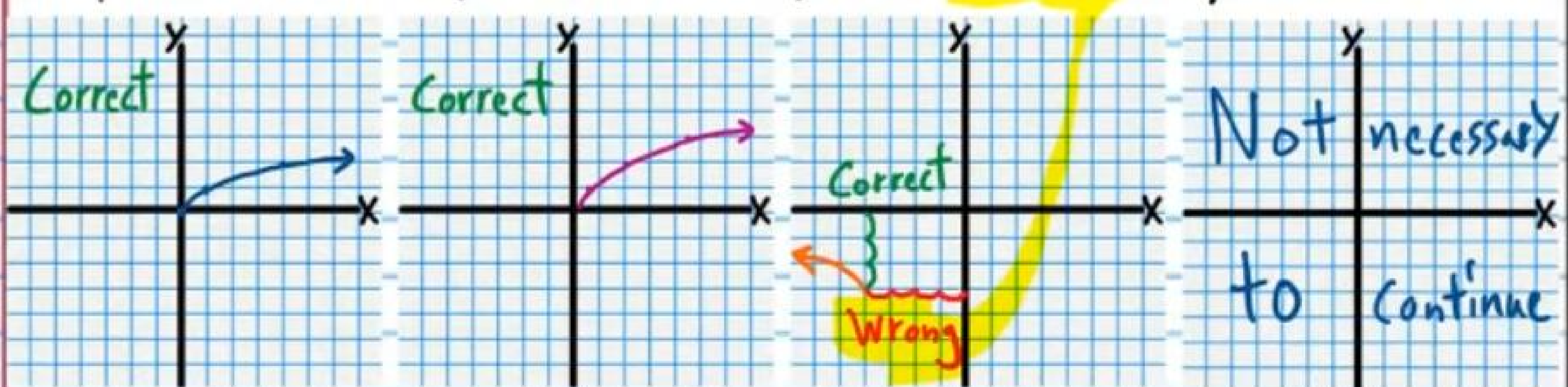
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Reflection	Translation?	Dilation
$y=\sqrt{x}$	$y=\sqrt{-x}$	$y+3=\sqrt{-x+4}$	$y+3=\sqrt{-2x+4}$



Why? When "x" is replaced with x+4 in an equation, the graph of the new is equation is 4 units to the left of the graph of the original equation. However, we did not replace "x" with x+4. We replaced −x with −x+4. If we replace "x" with x+4, we would get $\sqrt{-(x+4)}$, which is equivalent to $\sqrt{-x-4}$, but −x−4 is not in the equation that we are trying to graph.

(1D) $y+3=\sqrt{-2x+4}$

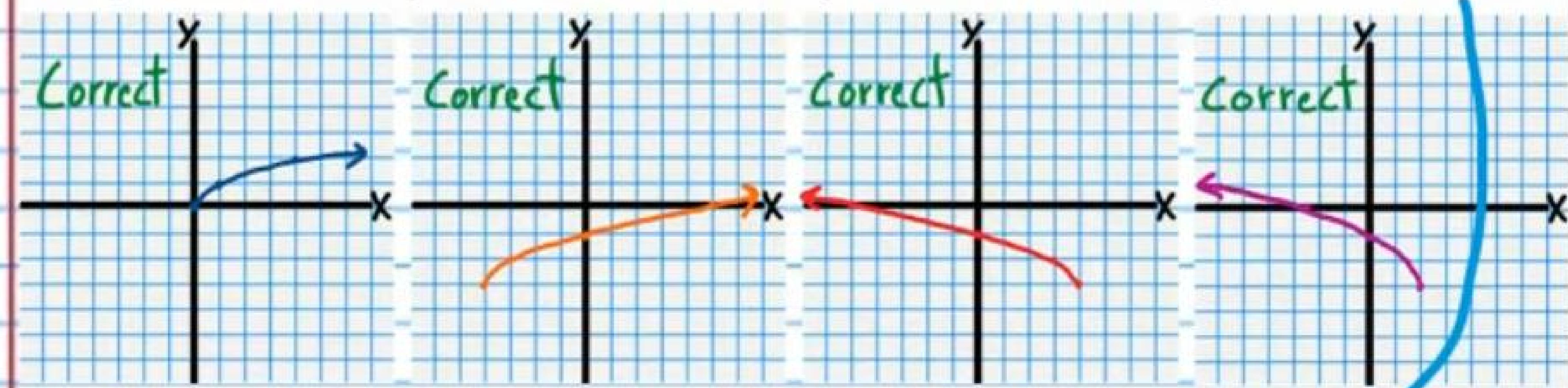
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Dilation	Translation?	Reflection
$y=\sqrt{x}$	$y=\sqrt{2x}$	$y+3=\sqrt{2x+4}$	$y+3=\sqrt{-2x+4}$



Examples Demonstrating Correct Usage of Transformations

(1E) $y+3=\sqrt{-2x+4}$

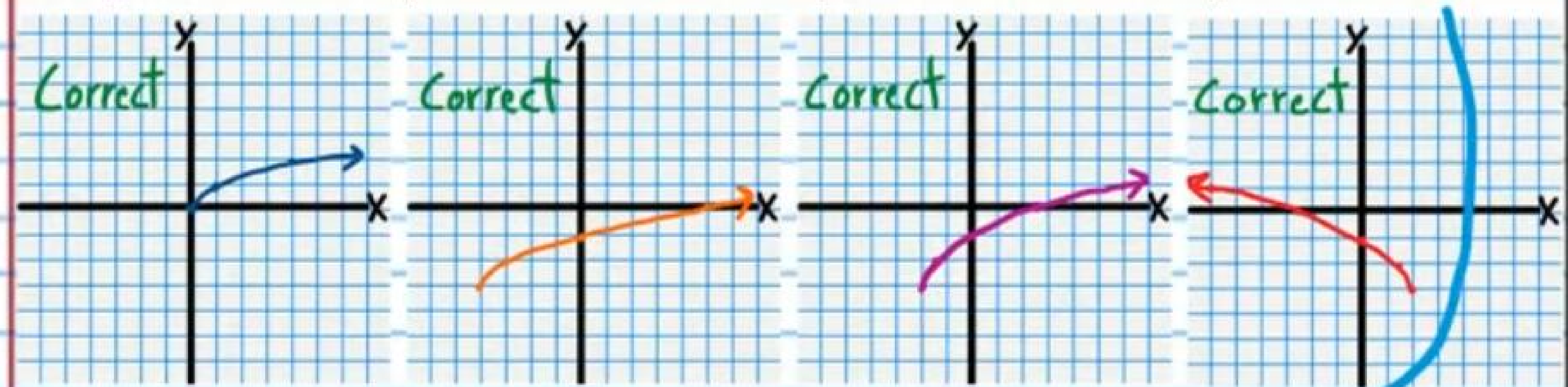
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Translation	Reflection	Dilation
$y=\sqrt{x}$	$y+3=\sqrt{x+4}$	$y+3=\sqrt{-x+4}$	$y+3=\sqrt{-(2x)+4}$



$-(2x)=-2x$

(1F) $y+3=\sqrt{-2x+4}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Translation	Dilation	Reflection
$y=\sqrt{x}$	$y+3=\sqrt{x+4}$	$y+3=\sqrt{2x+4}$	$y+3=\sqrt{2(-x)+4}$

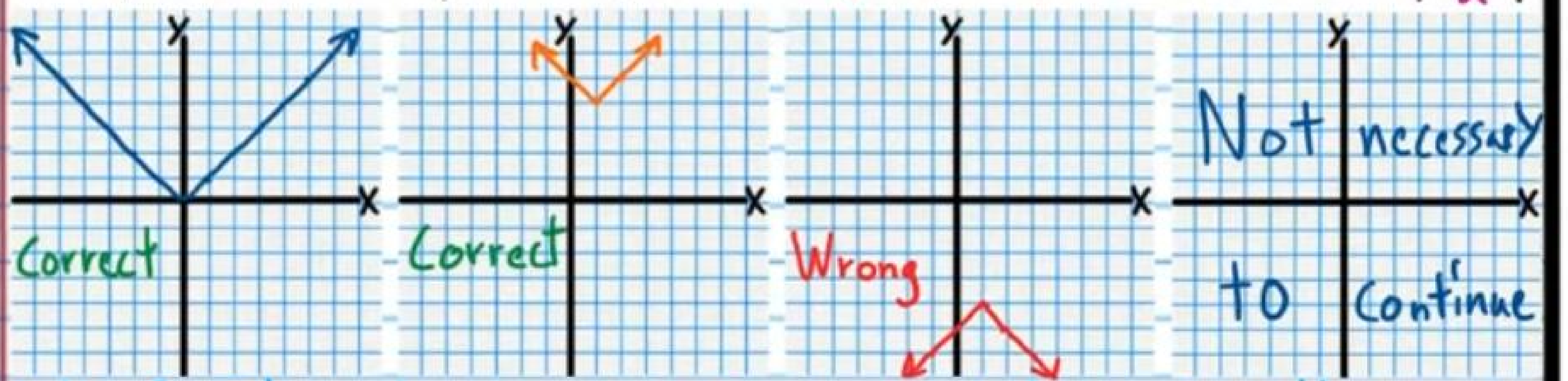


$2(-x)=-2x.$

Examples Demonstrating Incorrect Usage of Transformations

2A $-(y-4)=\left|\frac{x-1}{2}\right|$

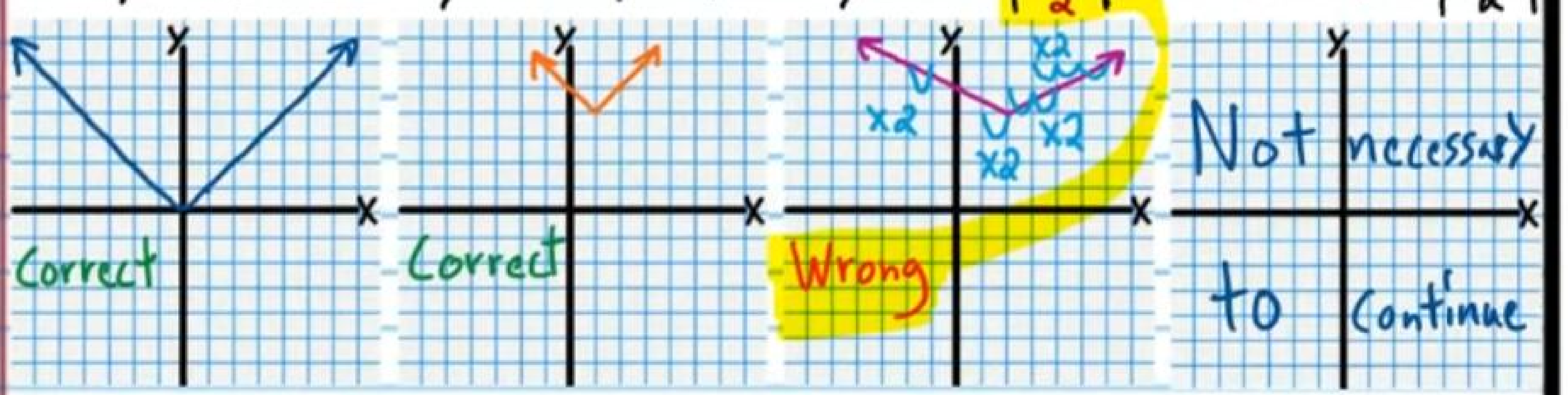
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Translation	Reflection?	Dilation?
$y=\lvert x\rvert$	$y-4=\lvert x-1\rvert$	$-(y-4)=\lvert x-1\rvert$	$-(y-4)=\left\lvert\frac{x-1}{2}\right\rvert$



Why? It is because in the second transformation we replaced $y-4$ with $-(y-4)$ rather than replacing y with $-y$. Our transformation properties do not apply properly when modifying an equation this way. Also, we replaced $x-1$ with $\frac{x-1}{2}$, rather than replacing x with $\frac{x}{2}$. The transformation properties cannot be applied properly.

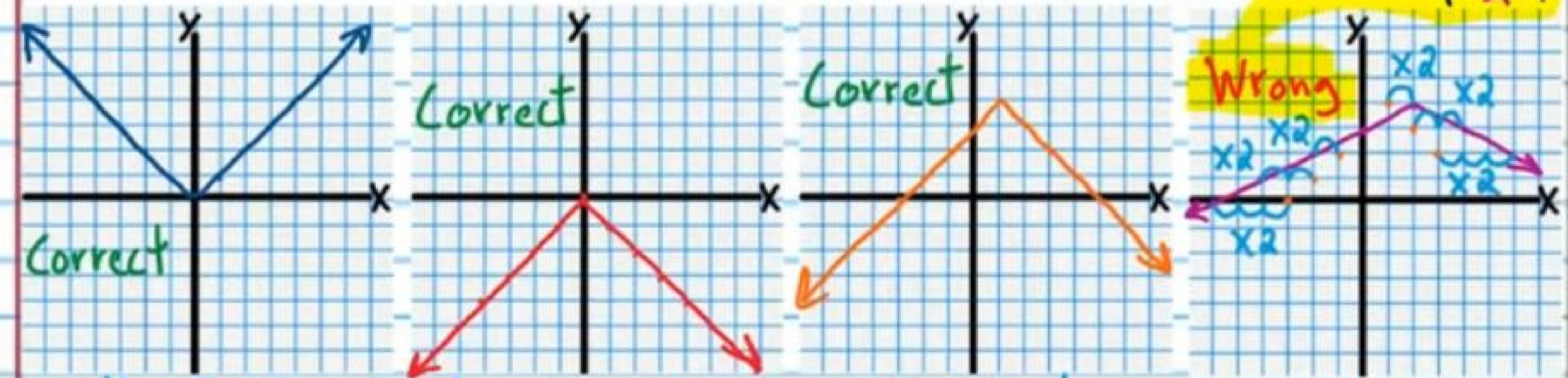
2B $-(y-4)=\left|\frac{x-1}{2}\right|$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Translation	Dilation?	Reflection?
$y=\lvert x\rvert$	$y-4=\lvert x-1\rvert$	$y-4=\left\lvert\frac{x-1}{2}\right\rvert$	$-(y-4)=\left\lvert\frac{x-1}{2}\right\rvert$



2C $-(y-4)=\left|\frac{x-1}{2}\right|$

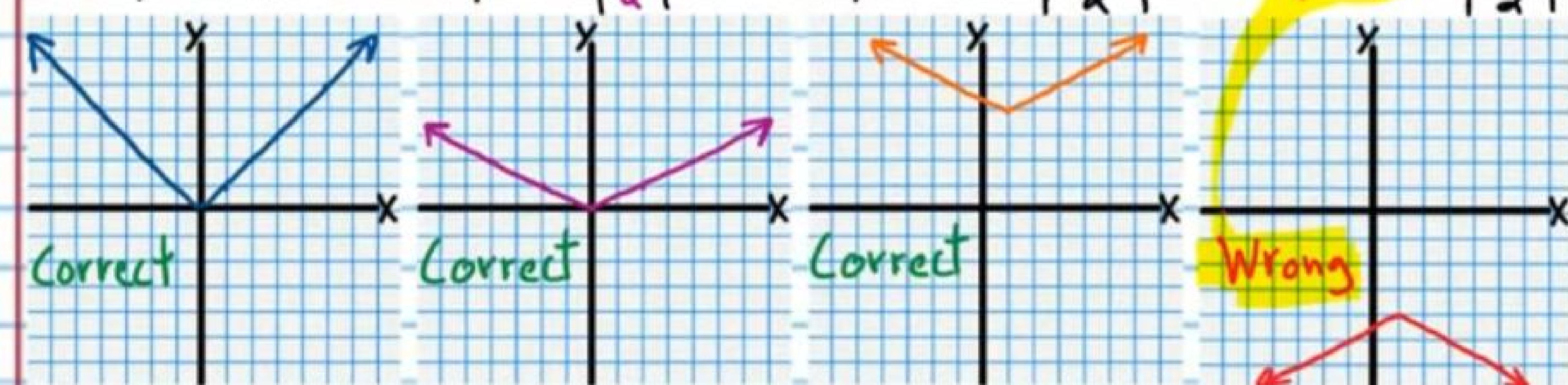
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Reflection	Translation	Dilation?
$y=\lvert x\rvert$	$-y=\lvert x\rvert$	$-(y-4)=\lvert x-1\rvert$	$-(y-4)=\left\lvert\frac{x-1}{2}\right\rvert$



Why? It is because in the third transformation we replaced $x-1$ with $\frac{x-1}{2}$ rather than replacing x with $\frac{x}{2}$. Our transformation properties do not apply properly when modifying an equation this way.

2D $-(y-4)=\left|\frac{x-1}{2}\right|$

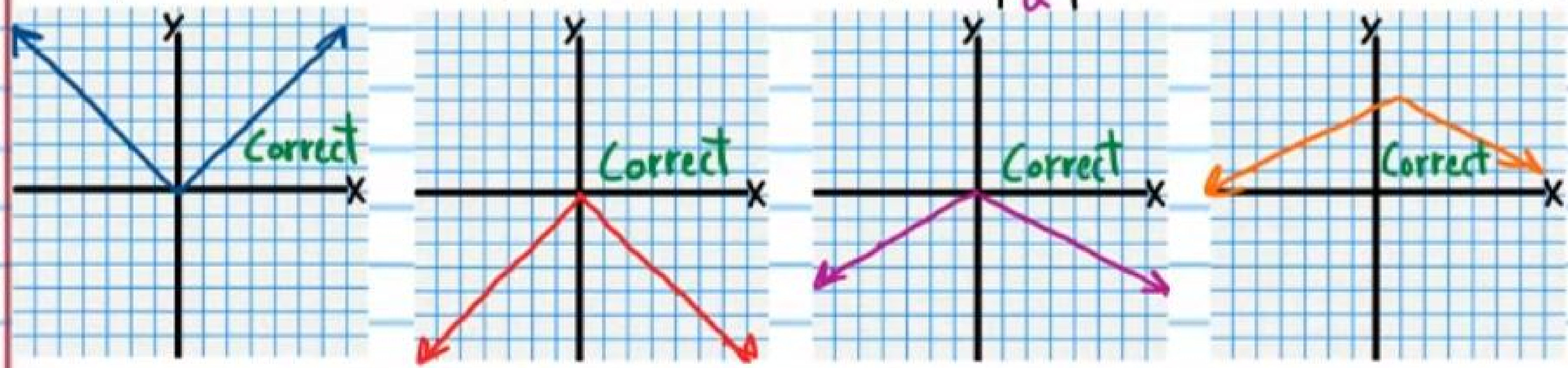
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Dilation	Translation	Reflection?
$y=\lvert x\rvert$	$y=\left\lvert\frac{x}{2}\right\rvert$	$y-4=\left\lvert\frac{x-1}{2}\right\rvert$	$-(y-4)=\left\lvert\frac{x-1}{2}\right\rvert$



Examples Demonstrating Correct Usage of Transformations

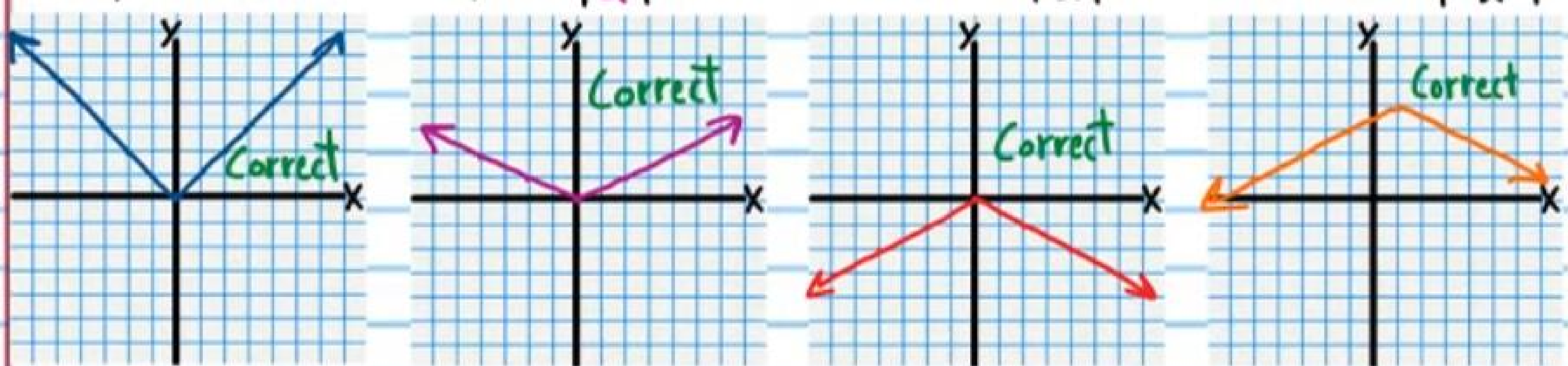
2E $-(y-4) = \left|\frac{x-1}{2}\right|$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Reflection	Dilation	Translation
$y=\lvert x\rvert$	$-y=\lvert x\rvert$	$-y=\left\lvert\frac{x}{2}\right\rvert$	$-(y-4)=\left\lvert\frac{x-1}{2}\right\rvert$



2F $-(y-4) = \left|\frac{x-1}{2}\right|$

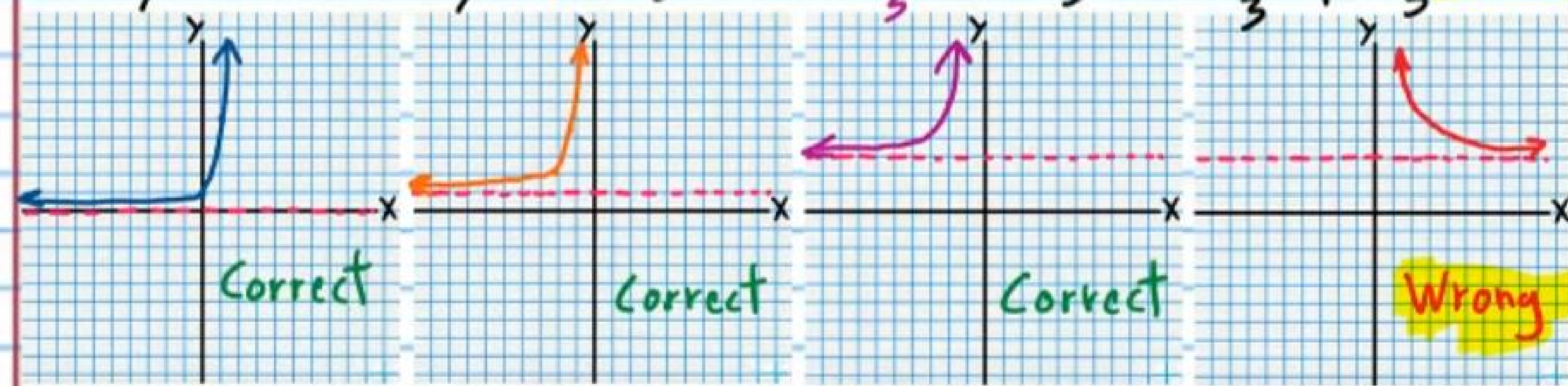
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Dilation	Reflection	Translation
$y=\lvert x\rvert$	$y=\left\lvert\frac{x}{2}\right\rvert$	$-y=\left\lvert\frac{x}{2}\right\rvert$	$-(y-4)=\left\lvert\frac{x-1}{2}\right\rvert$



Examples Demonstrating Incorrect Usage of Transformations

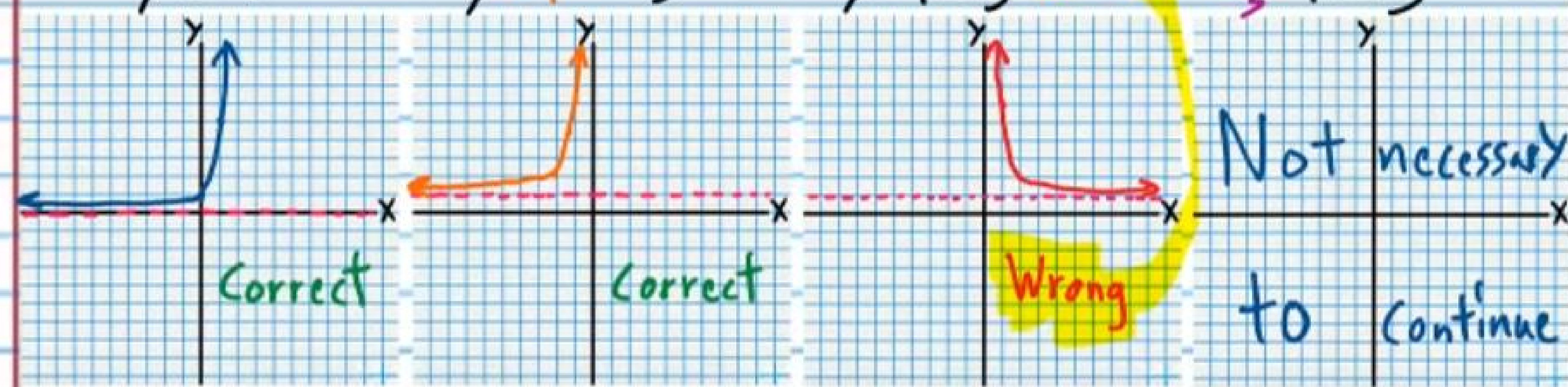
3A $\frac{y}{3}-1 = 5^{-(x+2)}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Translation	Dilation	Reflection?
$y=5^x$	$y-1=5^{x+2}$	$\frac{y}{3}-1=5^{x+2}$	$\frac{y}{3}-1=5^{-(x+2)}$



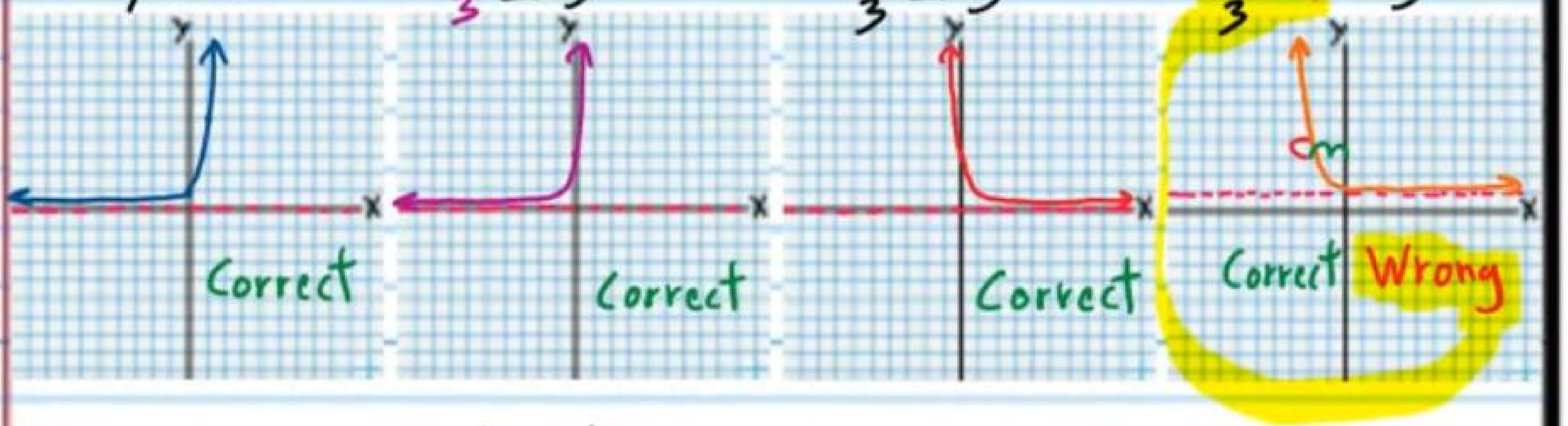
3B $\frac{y}{3}-1 = 5^{-(x+2)}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Translation	Reflection?	Dilation
$y=5^x$	$y-1=5^{x+2}$	$y-1=5^{-(x+2)}$	$\frac{y}{3}-1=5^{-(x+2)}$



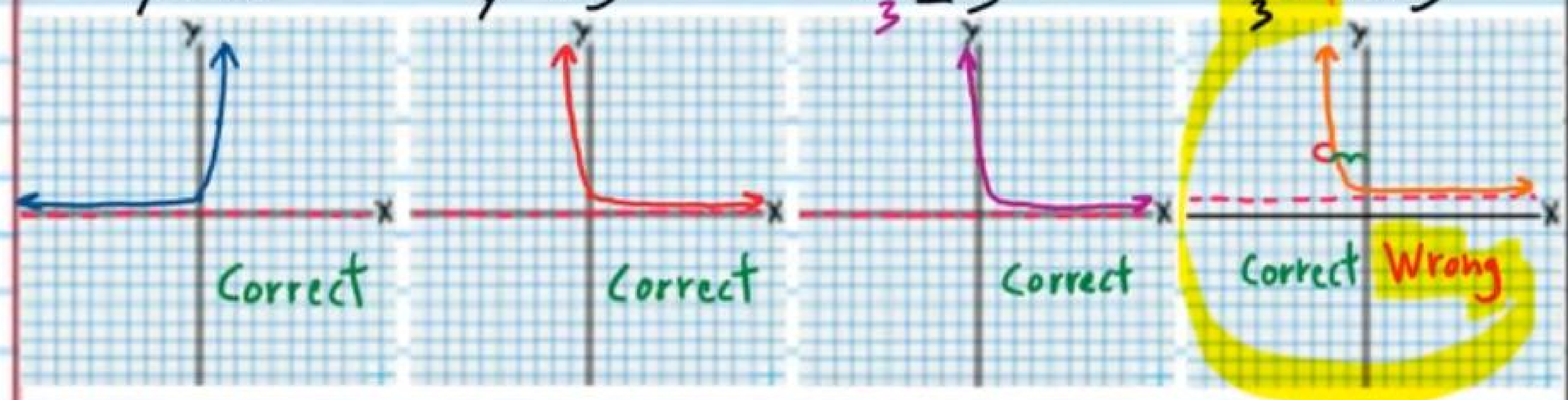
(3C) $\frac{y}{3}-1=5^{-(x+2)}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Dilation	Reflection	Translation?
$y=5^x$	$\frac{y}{3}=5^x$	$\frac{y}{3}=5^{-x}$	$\frac{y}{3}-1=5^{-(x+2)}$



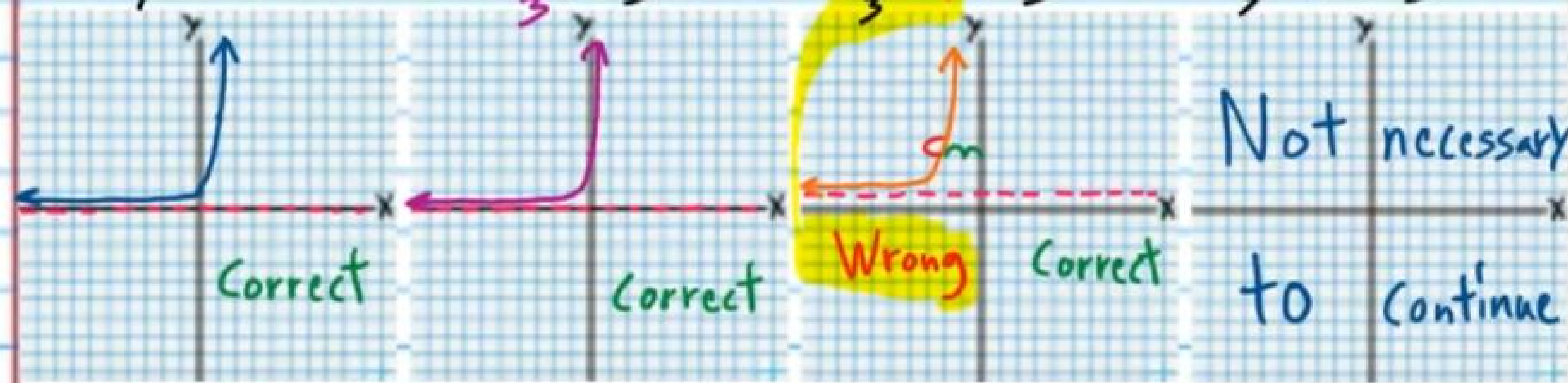
(3D) $\frac{y}{3}-1=5^{-(x+2)}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Reflection	Dilation	Translation?
$y=5^x$	$y=5^{-x}$	$\frac{y}{3}=5^{-x}$	$\frac{y}{3}-1=5^{-(x+2)}$



(3E) $\frac{y}{3}-1=5^{-(x+2)}$

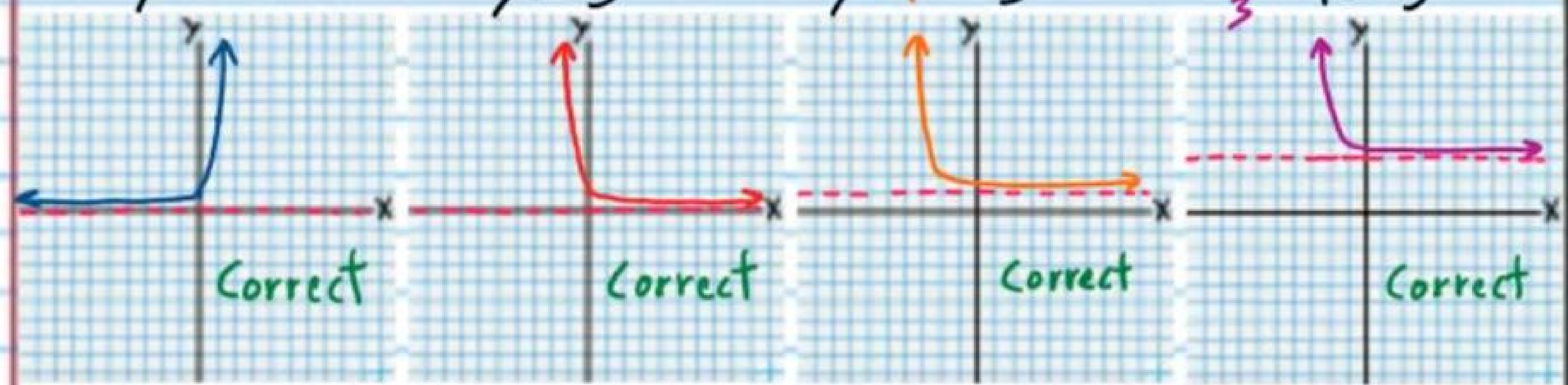
Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Dilation	Translation?	Reflection?
$y=5^x$	$\frac{y}{3}=5^x$	$\frac{y}{3}-1=5^{x+2}$	$\frac{y}{3}-1=5^{-(x+2)}$



An Example Demonstrating Correct Usage of Transformations

(3F) $\frac{y}{3}-1=5^{-(x+2)}$

Parent Relation	Transformation #1	Transformation #2	Transformation #3
	Reflection	Translation	Dilation
$y=5^x$	$y=5^{-x}$	$y-1=5^{-(x+2)}$	$\frac{y}{3}-1=5^{-(x+2)}$



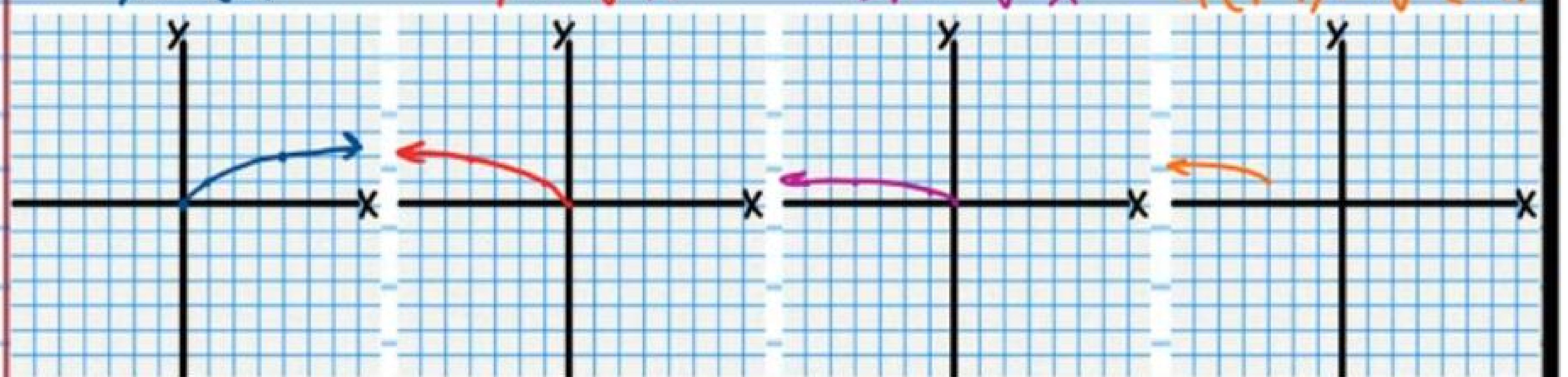
The transformation properties can be used with each pathway below to graph the corresponding equations. But in most of these pathways, expressions must be converted into a different form in order to use the transformation properties in the way that they are written. Choose the pathways in which no modification of expressions is required. Then, use your choosen pathway to graph the corresponding equation.

① $2(y-1)=\sqrt{-(x+3)}$

Pathway #1	Pathway #2	Pathway #3
$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓
$y-1=\sqrt{x+3}$ ✓	$y-1=\sqrt{x+3}$ ✓	$2y=\sqrt{x}$ ✓
$2(y-1)=\sqrt{x+3}$ ✗	$y-1=\sqrt{-(x+3)}$ ✗	$2y=\sqrt{-x}$ ✓
$2(y-1)=\sqrt{-(x+3)}$ ✗	$2(y-1)=\sqrt{-(x+3)}$ ✗	$2(y-1)=\sqrt{-(x+3)}$ ✓

Pathway #4	Pathway #5	Pathway #6
$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓	$y=\sqrt{x}$ ✓
$2y=\sqrt{x}$ ✓	$y=\sqrt{-x}$ ✓	$y=\sqrt{-x}$ ✓
$2(y-1)=\sqrt{x+3}$ ✓	$2y=\sqrt{-x}$ ✓	$y-1=\sqrt{-(x+3)}$ ✓
$2(y-1)=\sqrt{-(x+3)}$ ✗	$2(y-1)=\sqrt{-(x+3)}$ ✓	$2(y-1)=\sqrt{-(x+3)}$ ✗

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Dilation	Translation
$y=\sqrt{x}$	$y=\sqrt{-x}$	$2y=\sqrt{-x}$	$2(y-1)=\sqrt{-(x+3)}$

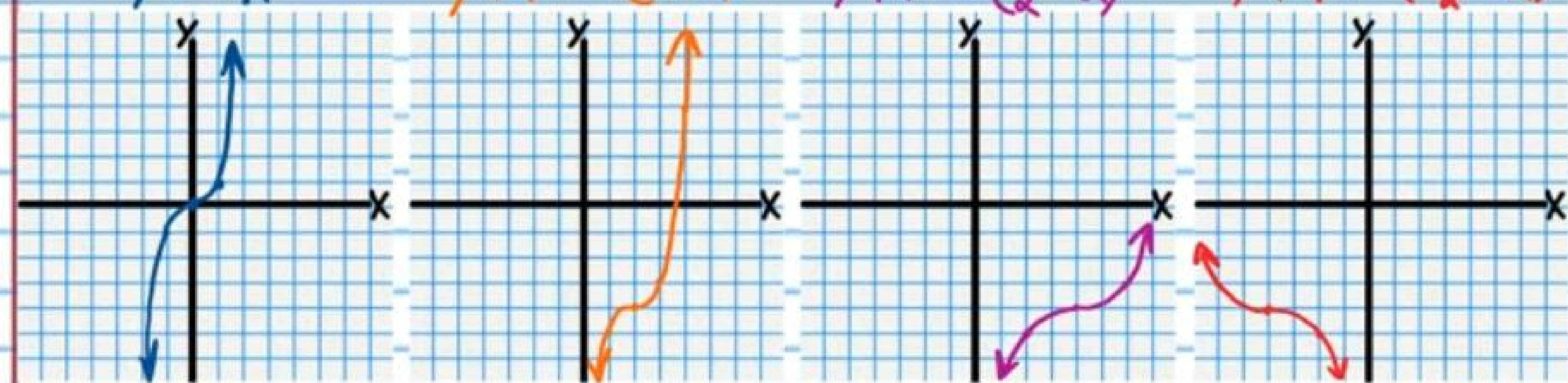


② $y+4=\left(-\frac{x}{2}-2\right)^3$

Pathway #1	Pathway #2	Pathway #3
$y=x^3$ ✓	$y=x^3$ ✓	$y=x^3$ ✓
$y+4=(x-2)^3$ ✓	$y+4=(x-2)^3$ ✓	$y=\left(\frac{x}{2}\right)^3$ ✓
$y+4=\left(\frac{x}{2}-2\right)^3$ ✓	$y+4=(-x-2)^3$ ✓	$y=\left(\frac{-x}{2}\right)^3$ ✓
$y+4=\left(\frac{-x}{2}-2\right)^3$ ✓	$y+4=\left(-\frac{x}{2}-2\right)^3$ ✓	$y+4=\left(-\frac{x}{2}-2\right)^3$ ✗

Pathway #4	Pathway #5	Pathway #6
$y=x^3$ ✓	$y=x^3$ ✓	$y=x^3$ ✓
$y=\left(\frac{x}{2}\right)^3$ ✓	$y=(-x)^3$ ✓	$y=(-x)^3$ ✓
$y+4=\left(\frac{x}{2}-2\right)^3$ ✗	$y=\left(-\frac{x}{2}\right)^3$ ✓	$y+4=(-x-2)^3$ ✗
$y+4=\left(\frac{-x}{2}-2\right)^3$ ✓	$y+4=\left(-\frac{x}{2}-2\right)^3$ ✗	$y+4=\left(-\frac{x}{2}-2\right)^3$ ✓

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Translation	Dilation	Reflection
$y=x^3$	$y+4=(x-2)^3$	$y+4=\left(\frac{x}{2}-2\right)^3$	$y+4=\left(\frac{-x}{2}-2\right)^3$



Logarithm Property III	Logarithm Property II	Logarithm Property IX
$\log_b b = 1$	$\log_b b^x = x$	$\log_b 1 = 0$

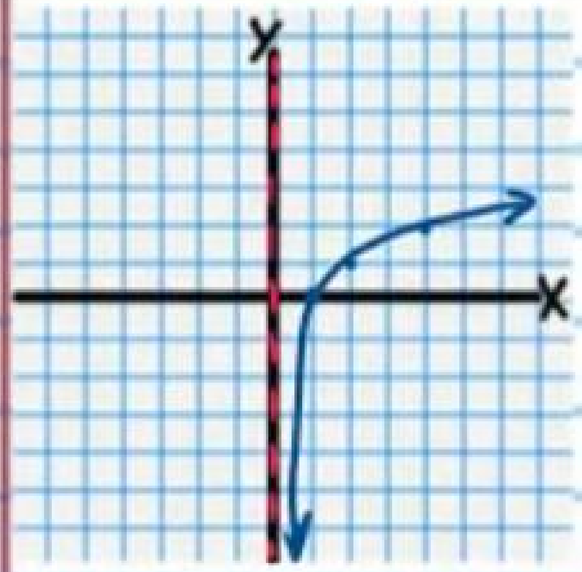
③ $-y-3=\log_2\left(\frac{x+1}{3}\right)$

Pathway #1	Pathway #2	Pathway #3
$y=\log_2 x$ ✓	$y=\log_2 x$ ✓	$y=\log_2 x$ ✓
$y-3=\log_2(x+1)$ ✓	$y-3=\log_2(x+1)$ ✓	$y=\log_2\left(\frac{x}{3}\right)$ ✓
$y-3=\log_2\left(\frac{x+1}{3}\right)$ ✗	$-y-3=\log_2(x+1)$ ✓	$-y=\log_2\left(\frac{x}{3}\right)$ ✓
$-y-3=\log_2\left(\frac{x+1}{3}\right)$ ✓	$-y-3=\log_2\left(\frac{x+1}{3}\right)$ ✗	$-y-3=\log_2\left(\frac{x+1}{3}\right)$ ✗

Pathway #4	Pathway #5	Pathway #6
$y=\log_2 x$ ✓	$y=\log_2 x$ ✓	$y=\log_2 x$ ✓
$y=\log_2\left(\frac{x}{3}\right)$ ✓	$-y=\log_2 x$ ✓	$-y=\log_2 x$ ✓
$y-3=\log_2\left(\frac{x+1}{3}\right)$ ✓	$-y=\log_2\left(\frac{x}{3}\right)$ ✓	$-y-3=\log_2(x+1)$ ✗
$-y-3=\log_2\left(\frac{x+1}{3}\right)$ ✓	$-y-3=\log_2\left(\frac{x+1}{3}\right)$ ✗	$-y-3=\log_2\left(\frac{x+1}{3}\right)$ ✗

Parent Relation

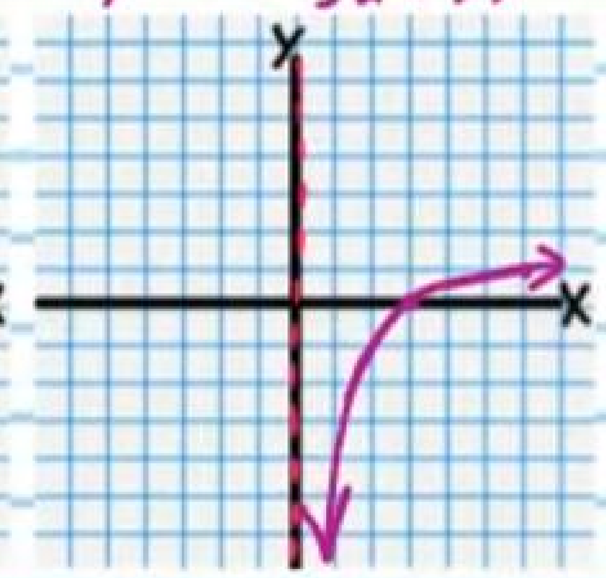
$y=\log_2 x$



Transformation #1

Dilation

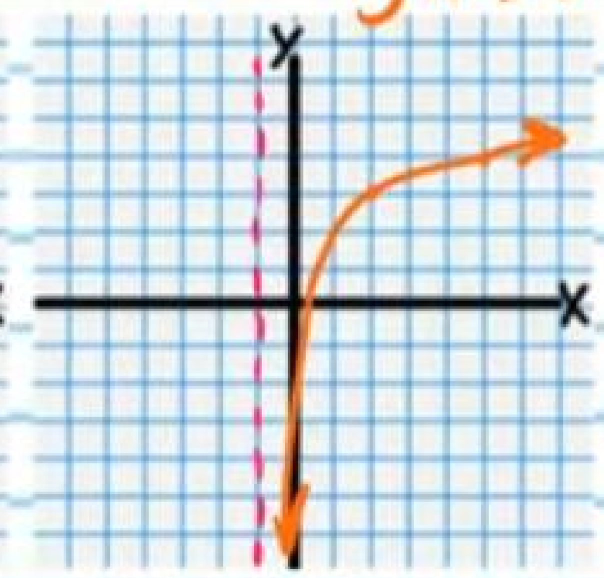
$y=\log_2\left(\frac{x}{3}\right)$



Transformation #2

Translation

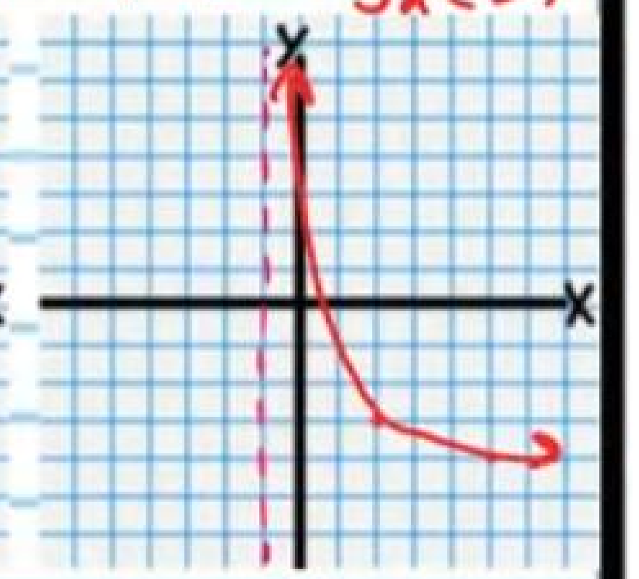
$y-3=\log_2\left(\frac{x+1}{3}\right)$



Final Transformation

Reflection

$-y-3=\log_2\left(\frac{x+1}{3}\right)$



$2(-x)=-2x$

④ $y+2=\frac{1}{-2x+3}$

Pathway #1	Pathway #2	Pathway #3
$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓
$y+2=\frac{1}{x+3}$ ✓	$y+2=\frac{1}{x+3}$ ✓	$y=\frac{1}{2x}$ ✓
$y+2=\frac{1}{2x+3}$ ✓	$y+2=\frac{1}{-x+3}$ ✓	$y=\frac{1}{2(-x)}$ ✓
$y+2=\frac{1}{2(-x)+3}$ ✓	$y+2=\frac{1}{-2x+3}$ ✓	$y+2=\frac{1}{-2x+3}$ ✗

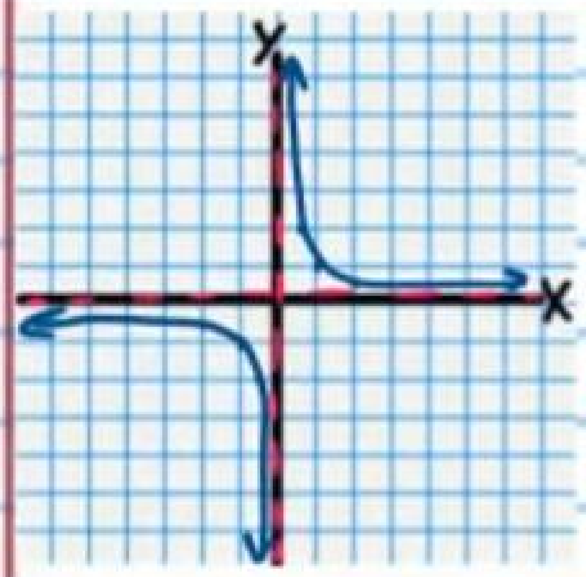
$2(-x)=-2x$

Pathway #4	Pathway #5	Pathway #6
$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓
$y=\frac{1}{2x}$ ✓	$y=\frac{1}{-x}$ ✓	$y=\frac{1}{-x}$ ✓
$y+2=\frac{1}{2x+3}$ ✗	$y=\frac{1}{-2x}$ ✓	$y+2=\frac{1}{-x+3}$ ✗
$y+2=\frac{1}{2(-x)+3}$ ✓	$y+2=\frac{1}{-2x+3}$ ✗	$y+2=\frac{1}{-2x+3}$ ✓

$2(-x)=-2x$

Parent Relation

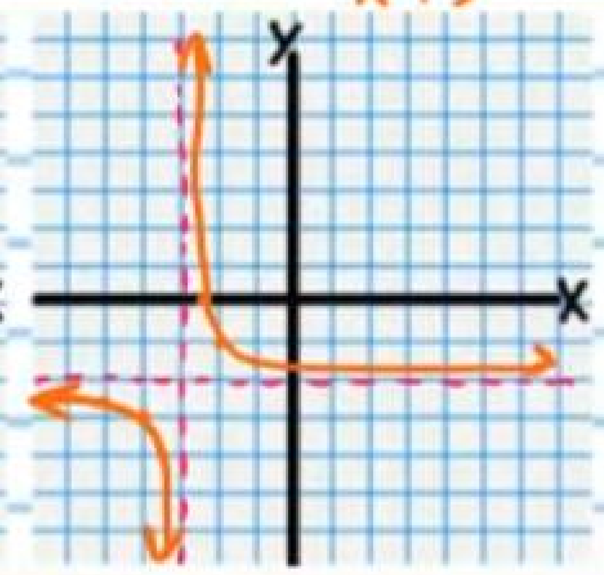
$y=\frac{1}{x}$



Transformation #1

Translation

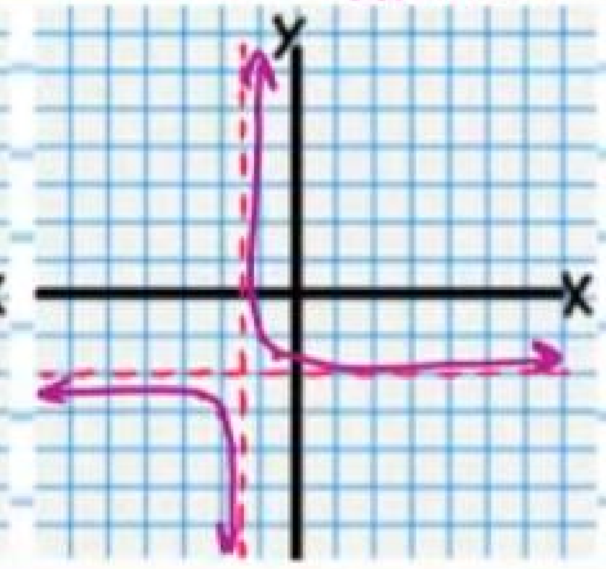
$y+2=\frac{1}{x+3}$



Transformation #2

Dilation

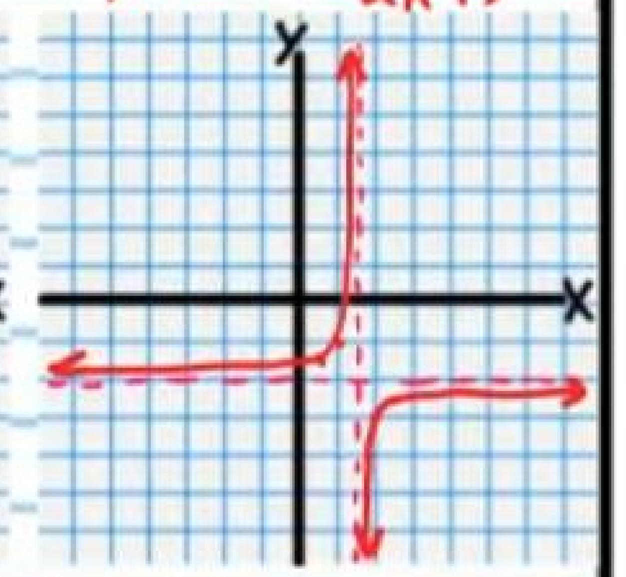
$y+2=\frac{1}{2x+3}$



Final Transformation

Reflection

$y+2=\frac{1}{-2x+3}$

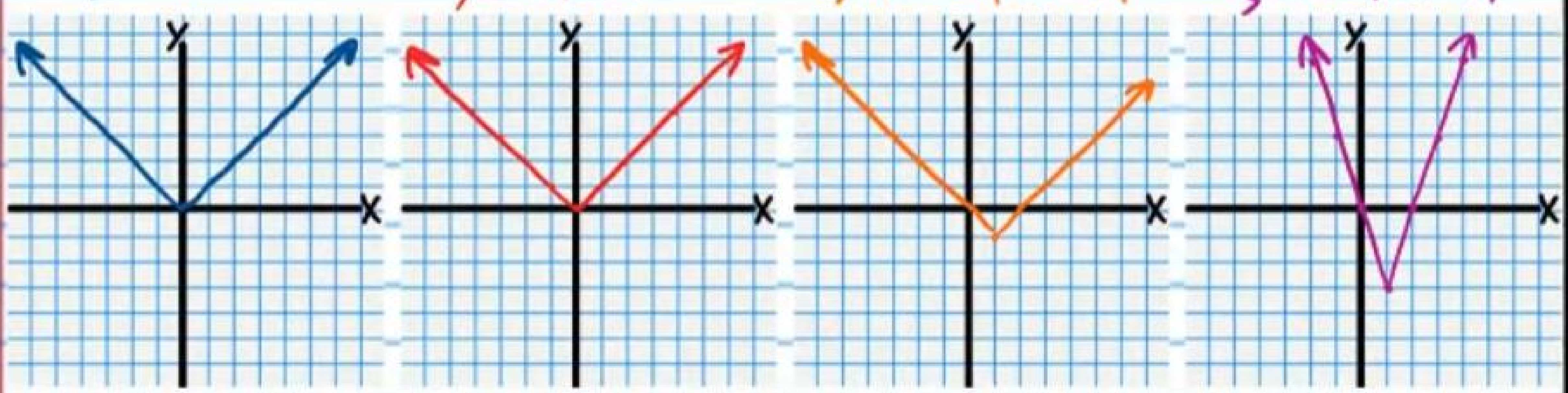


⑤ $\frac{y}{3}+1=|-(x-1)|$

Pathway #1	Pathway #2	Pathway #3
$y=\|x\|$ ✓	$y=\|x\|$ ✓	$y=\|x\|$ ✓
$y+1=\|x-1\|$ ✓	$y+1=\|x-1\|$ ✓	$\frac{y}{3}=\|x\|$ ✓
$\frac{y}{3}+1=\|x-1\|$ ✓	$y+1=\|-(x-1)\|$ ✗	$\frac{y}{3}+1=\|x-1\|$ ✗
$\frac{y}{3}+1=\|-(x-1)\|$ ✗	$\frac{y}{3}+1=\|-(x-1)\|$ ✓	$\frac{y}{3}+1=\|-(x-1)\|$ ✗

Pathway #4	Pathway #5	Pathway #6
$y=\|x\|$ ✓	$y=\|x\|$ ✓	$y=\|x\|$ ✓
$\frac{y}{3}=\|x\|$ ✓	$y=\|-x\|$ ✓	$y=\|-x\|$ ✓
$\frac{y}{3}=\|-x\|$ ✓	$\frac{y}{3}=\|-x\|$ ✓	$y+1=\|-(x-1)\|$ ✓
$\frac{y}{3}+1=\|-(x-1)\|$ ✗	$\frac{y}{3}+1=\|-(x-1)\|$ ✗	$\frac{y}{3}+1=\|-(x-1)\|$ ✓

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Translation	Dilation
$y=\|x\|$	$y=\|-x\|$	$y+1=\|-(x-1)\|$	$\frac{y}{3}+1=\|-(x-1)\|$

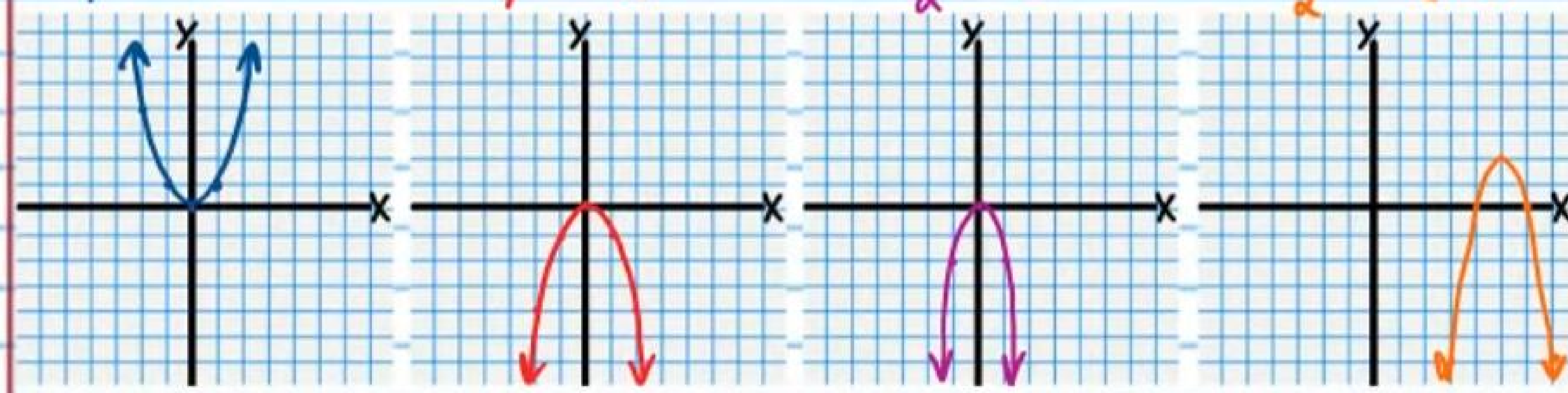


⑥ $\frac{-(y-2)}{2}=(x-5)^2$

Pathway #1	Pathway #2	Pathway #3
$y=x^2$ ✓	$y=x^2$ ✓	$y=x^2$ ✓
$y-2=(x-5)^2$ ✓	$y-2=(x-5)^2$ ✓	$\frac{y}{2}=x^2$ ✓
$\frac{y-2}{2}=(x-5)^2$ ✗	$-(y-2)=(x-5)^2$ ✗	$\frac{-y}{2}=x^2$ ✓
$\frac{-(y-2)}{2}=(x-5)^2$ ✗	$\frac{-(y-2)}{2}=(x-5)^2$ ✗	$\frac{-(y-2)}{2}=(x-5)^2$ ✓

Pathway #4	Pathway #5	Pathway #6
$y=x^2$ ✓	$y=x^2$ ✓	$y=x^2$ ✓
$\frac{y}{2}=x^2$ ✓	$-y=x^2$ ✓	$-y=x^2$ ✓
$\frac{y-2}{2}=(x-5)^2$ ✓	$-\frac{y}{2}=x^2$ ✓	$-(y-2)=(x-5)^2$ ✓
$\frac{-(y-2)}{2}=(x-5)^2$ ✗	$-\frac{y-2}{2}=(x-5)^2$ ✓	$\frac{-(y-2)}{2}=(x-5)^2$ ✗

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Dilation	Translation
$y=x^2$	$-y=x^2$	$-\frac{y}{2}=x^2$	$-\frac{y-2}{2}=(x-5)^2$



$2(-x)=-2x$

⑦ $y+1=2^{-2x+3}$

Pathway #1	Pathway #2	Pathway #3
$y=2^x$ ✓	$y=2^x$ ✓	$y=2^x$ ✓
$y+1=2^{x+3}$ ✓	$y+1=2^{x+3}$ ✓	$y=2^{2x}$ ✓
$y+1=2^{2x+3}$ ✓	$y+1=2^{-x+3}$ ✓	$y=2^{2(-x)}$ ✓
$y+1=2^{2(-x)+3}$ ✓	$y+1=2^{-2x+3}$ ✓	$y+1=2^{-2x+3}$ ✗

Pathway #4	Pathway #5	Pathway #6
$y=2^x$ ✓	$y=2^x$ ✓	$y=2^x$ ✓
$y=2^{2x}$ ✓	$y=2^{-x}$ ✓	$y=2^{-x}$ ✓
$y+1=2^{2x+3}$ ✗	$y=2^{-2x}$ ✓	$y+1=2^{-x+3}$ ✗
$y+1=2^{2(-x)+3}$ ✓	$y+1=2^{-2x+3}$ ✗	$y+1=2^{-2x+3}$ ✓

$2(-x)=-2x$

$2(-x)=-2x$

$2(-x)=-2x$

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Translation	Dilation	Reflection
$y=2^x$	$y+1=2^{x+3}$	$y+1=2^{2x+3}$	$y+1=2^{-2x+3}$

⑧ $\dfrac{-(y-1)}{3}=\dfrac{1}{x-1}$

Pathway #1	Pathway #2	Pathway #3
$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓
$y-1=\frac{1}{x-1}$ ✓	$y-1=\frac{1}{x-1}$ ✓	$\frac{y}{3}=\frac{1}{x}$ ✓
$\frac{y-1}{3}=\frac{1}{x-1}$ ✗	$-(y-1)=\frac{1}{x-1}$ ✗	$\frac{-y}{3}=\frac{1}{x}$ ✓
$\frac{-(y-1)}{3}=\frac{1}{x-1}$ ✗	$\frac{-(y-1)}{3}=\frac{1}{x-1}$ ✗	$\frac{-(y-1)}{3}=\frac{1}{x-1}$ ✓

Pathway #4	Pathway #5	Pathway #6
$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓	$y=\frac{1}{x}$ ✓
$\frac{y}{3}=\frac{1}{x}$ ✓	$-y=\frac{1}{x}$ ✓	$-y=\frac{1}{x}$ ✓
$\frac{y-1}{3}=\frac{1}{x-1}$ ✓	$-\frac{y}{3}=\frac{1}{x}$ ✓	$-(y-1)=\frac{1}{x-1}$ ✓
$\frac{-(y-1)}{3}=\frac{1}{x-1}$ ✗	$-\frac{y-1}{3}=\frac{1}{x-1}$ ✓	$\frac{-(y-1)}{3}=\frac{1}{x-1}$ ✗

Parent Relation	Transformation #1	Transformation #2	Final Transformation
	Reflection	Dilation	Translation
$y=\frac{1}{x}$	$-y=\frac{1}{x}$	$-\frac{y}{3}=\frac{1}{x}$	$-\frac{y-1}{3}=\frac{1}{x-1}$